

AI exposure, wage, and the comparative-advantage interaction: private-sector employment and hiring, by age group with within-firm (firm-FE) robustness

AI-Norway

June 10, 2026

This note collects the comparative-advantage (CA) results for two outcomes: *count* (cell employment headcount) and *nyjobb* (new hires = round(count × ny_jobb), cells with count ≥ 10). The full model interacts standardized exposure $z(\text{exp})$, standardized log full-time-equivalent wage $z(\text{lnw})$, and their product with event-time (figures) or with Post = $\mathbf{1}\{k \geq 0\}$ (tables). Poisson PPML, occupation + month fixed effects, SE clustered at occupation, private sector.

Why this interaction. Standard exposure measures capture AI’s *absolute* advantage in a task — whether AI can plausibly perform it. Lindenlaub, Oh, Rodríguez & Veldkamp (2026, henceforth LORV) emphasize that actual adoption is governed by AI’s *comparative* advantage relative to the worker performing the task. In their model, AI replaces labor in task k when

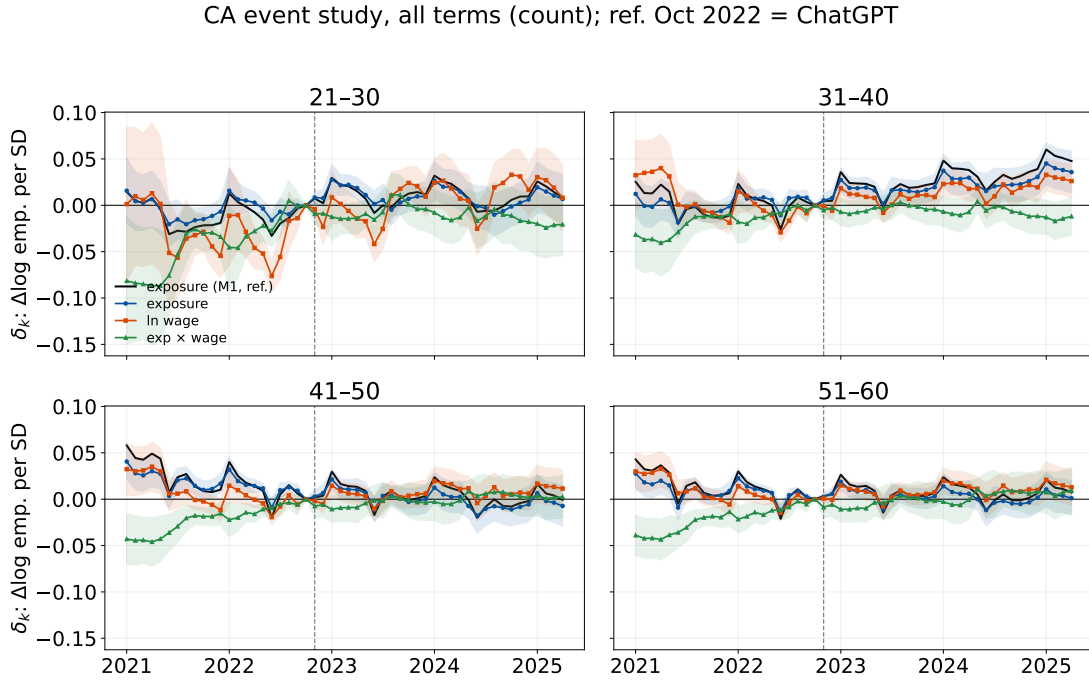
$$\underbrace{\frac{Z_{ik}^{\text{worker}}}{w_{io}}}_{\text{worker productivity per \$}} < \underbrace{\frac{Z_k^{\text{AI}}}{r_k}}_{\text{AI productivity per \$}}, \quad (1)$$

where Z_{ik}^{worker} , Z_k^{AI} are worker and AI productivity in task k , w_{io} is the worker’s wage, and r_k is the user cost of AI in task k . Exposure indices proxy the numerator on the right (AI’s productivity Z_k^{AI}); the wage w_{io} enters the denominator on the left. Both shifts move the inequality toward AI adoption, so the **combination** of high exposure **and** high wage creates the strongest incentive to actually automate: AI’s productivity gain is large and the labor it would replace is expensive. In a representative German linked employer–employee dataset LORV show that this comparative-advantage logic predicts roughly 60% of cross-occupation variation in observed AI adoption, against only 14% for an exposure-only model.

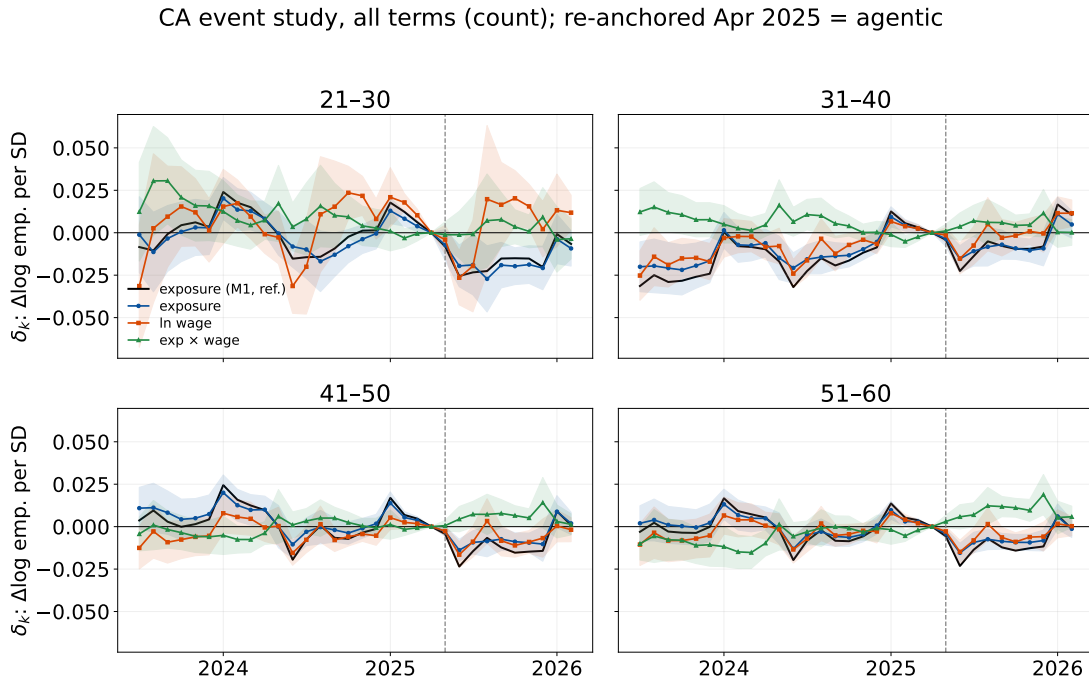
This is the structural rationale for our interaction term $z(\text{exp}) \times z(\text{lnw})$: it captures the comparative-advantage hotspots where adoption (and therefore any labor-market displacement) should be concentrated. Read in this light, *M1* (exposure only) measures absolute advantage; *M2* (wage only) is the labor-side cost margin; *M3* (full) adds the comparative-advantage margin. The event-time figures plot the same three paths over k , with a black reference line for *M1* to make the comparative-advantage margin visible.

1 Event studies (Poisson PPML, all terms)

1.1 Employment (count)



Private sector. Full model: $z(\text{exposure})$, $z(\ln \text{ FTE wage})$ and their interaction, each $\times$ event-time dummies ($k=-1$ ref). Black line = exposure estimated alone (M1). Poisson PPML, occ + month FE; 95% CI bands; SE clustered at occupation. Dashed = reference launch.

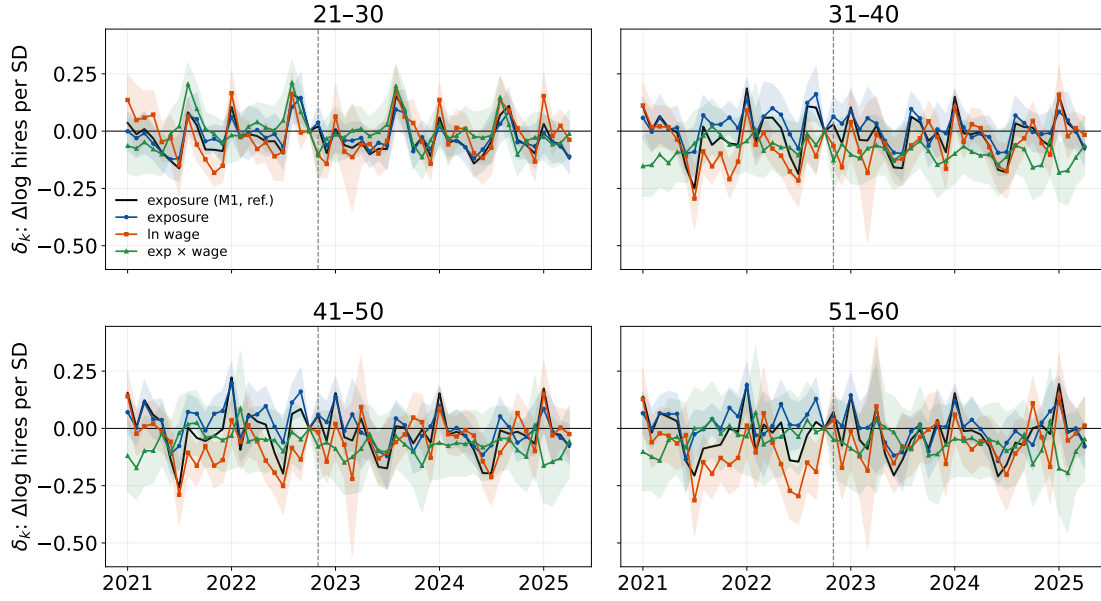


Private sector. Full model: $z(\text{exposure})$, $z(\ln \text{ FTE wage})$ and their interaction, each $\times$ event-time dummies ($k=-1$ ref). Black line = exposure estimated alone (M1). Poisson PPML, occ + month FE; 95% CI bands; SE clustered at occupation. Dashed = reference launch.

Figure 1: Employment (count): CA event study — ChatGPT anchor (top), agentic anchor (bottom).

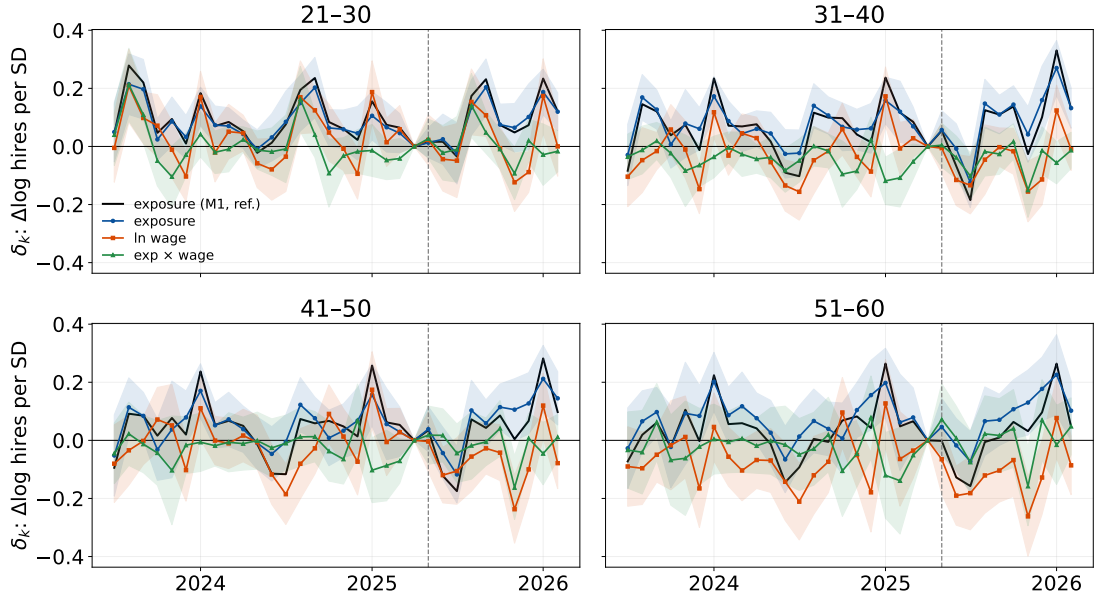
1.2 New hires (nyjobb)

CA event study, all terms (nyjobb); ref. Oct 2022 = ChatGPT



Private sector. Full model: $z(\text{exposure})$, $z(\ln \text{ FTE wage})$ and their interaction, each $\times$ event-time dummies ($k=-1$ ref). Black line = exposure estimated alone (M1). Poisson PPML, occ + month FE; 95% CI bands; SE clustered at occupation. Dashed = reference launch.

CA event study, all terms (nyjobb); re-anchored Apr 2025 = agentic



Private sector. Full model: $z(\text{exposure})$, $z(\ln \text{ FTE wage})$ and their interaction, each $\times$ event-time dummies ($k=-1$ ref). Black line = exposure estimated alone (M1). Poisson PPML, occ + month FE; 95% CI bands; SE clustered at occupation. Dashed = reference launch.

Figure 2: New hires (nyjobb): CA event study — ChatGPT anchor (top), agentic anchor (bottom).

2 Pooled post-treatment estimates (Poisson PPML DiD)

2.1 Employment (count)

Table 1: AI exposure, wage, and private-sector employment headcount, by age group

	ChatGPT (ref. Oct 2022)			Agentic (ref. Apr 2025)		
	(1)	(2)	(3)	(4)	(5)	(6)
<i>Panel A: ages 21–30</i>						
Exposure $\times$ Post	+0.0104 (0.0070)		+0.0090 (0.0080)	-0.0154*** (0.0053)		-0.0165*** (0.0054)
ln wage $\times$ Post		+0.0116 (0.0085)	+0.0068 (0.0101)		-0.0004 (0.0084)	+0.0048 (0.0083)
Exp $\times$ wage $\times$ Post			-0.0104 (0.0106)			+0.0018 (0.0058)
Observations		20,176			12,320	
Occupations		388			385	
<i>Panel B: ages 31–40</i>						
Exposure $\times$ Post	+0.0291*** (0.0061)		+0.0225*** (0.0072)	-0.0054* (0.0031)		-0.0059* (0.0034)
ln wage $\times$ Post		+0.0256*** (0.0053)	+0.0142** (0.0056)		-0.0026 (0.0036)	-0.0001 (0.0039)
Exp $\times$ wage $\times$ Post			-0.0060 (0.0071)			+0.0046 (0.0035)
Observations		20,072			12,320	
Occupations		386			385	
<i>Panel C: ages 41–50</i>						
Exposure $\times$ Post	+0.0041 (0.0060)		+0.0000 (0.0072)	-0.0096*** (0.0023)		-0.0060** (0.0026)
ln wage $\times$ Post		+0.0070 (0.0057)	+0.0070 (0.0067)		-0.0090*** (0.0029)	-0.0061* (0.0033)
Exp $\times$ wage $\times$ Post			-0.0008 (0.0086)			+0.0058* (0.0030)
Observations		20,124			12,320	
Occupations		387			385	
<i>Panel D: ages 51–60</i>						
Exposure $\times$ Post	+0.0076 (0.0054)		+0.0030 (0.0060)	-0.0095*** (0.0019)		-0.0067*** (0.0019)
ln wage $\times$ Post		+0.0108** (0.0053)	+0.0092 (0.0060)		-0.0074*** (0.0021)	-0.0050** (0.0022)
Exp $\times$ wage $\times$ Post			+0.0005 (0.0088)			+0.0091*** (0.0023)
Observations		20,176			12,352	
Occupations		388			386	
Occupation FE	Yes	Yes	Yes	Yes	Yes	Yes
Month FE	Yes	Yes	Yes	Yes	Yes	Yes

Notes: Poisson PPML (cell-level employment headcount, private sector). Columns (1)–(3): ChatGPT window (Oct 2022 = $k=-1$, through 2025m4); (4)–(6): agentic (Apr 2025 = $k=-1$, from 2023m7). Each coefficient is the average post-period effect relative to the reference month ($k=-1$); each pre-period month enters as a separate control. M1 = exposure only, M2 = ln wage only, M3 = both + interaction. Treatments standardized: $z(\text{exp}) = \text{Eloundou } \beta$ (pooled), $z(\text{lnw}) = \ln$ full-time-equivalent wage within age group. Occupation and month fixed effects in all columns; SE clustered at occupation in parentheses; coefficients are $\Delta \log$ per 1 SD. Pseudo $R^2 \approx 0.99$ (FE-driven). * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

2.2 New hires (nyjobb)

Table 2: AI exposure, wage, and private-sector new hires = round(count $\times$ ny_jobb), by age group

	ChatGPT (ref. Oct 2022)			Agentic (ref. Apr 2025)		
	(1)	(2)	(3)	(4)	(5)	(6)
<i>Panel A: ages 21–30</i>						
Exposure $\times$ Post	-0.0264 (0.0264)		-0.0240 (0.0282)	+0.1014*** (0.0266)		+0.0971*** (0.0289)
ln wage $\times$ Post		-0.0166 (0.0233)	-0.0089 (0.0261)		+0.0494 (0.0329)	+0.0230 (0.0327)
Exp $\times$ wage $\times$ Post			-0.0007 (0.0371)			+0.0151 (0.0257)
Observations		19,136			11,744	
Occupations		368			367	
<i>Panel B: ages 31–40</i>						
Exposure $\times$ Post	-0.0074 (0.0359)		+0.0121 (0.0323)	+0.0885*** (0.0232)		+0.1071*** (0.0269)
ln wage $\times$ Post		-0.0293 (0.0362)	-0.0352 (0.0329)		+0.0186 (0.0265)	-0.0368 (0.0278)
Exp $\times$ wage $\times$ Post			-0.0978* (0.0562)			-0.0284 (0.0229)
Observations		19,188			11,648	
Occupations		369			364	
<i>Panel C: ages 41–50</i>						
Exposure $\times$ Post	-0.0126 (0.0365)		-0.0013 (0.0280)	+0.0558*** (0.0214)		+0.0874** (0.0343)
ln wage $\times$ Post		-0.0295 (0.0392)	-0.0292 (0.0386)		-0.0074 (0.0294)	-0.0553 (0.0385)
Exp $\times$ wage $\times$ Post			-0.0802 (0.0661)			-0.0117 (0.0339)
Observations		18,980			11,616	
Occupations		365			363	
<i>Panel D: ages 51–60</i>						
Exposure $\times$ Post	-0.0069 (0.0371)		+0.0077 (0.0290)	+0.0438 (0.0284)		+0.1011** (0.0411)
ln wage $\times$ Post		-0.0373 (0.0467)	-0.0395 (0.0509)		-0.0490 (0.0402)	-0.1005** (0.0481)
Exp $\times$ wage $\times$ Post			-0.0653 (0.0742)			+0.0094 (0.0485)
Observations		18,980			11,392	
Occupations		365			356	
Occupation FE	Yes	Yes	Yes	Yes	Yes	Yes
Month FE	Yes	Yes	Yes	Yes	Yes	Yes

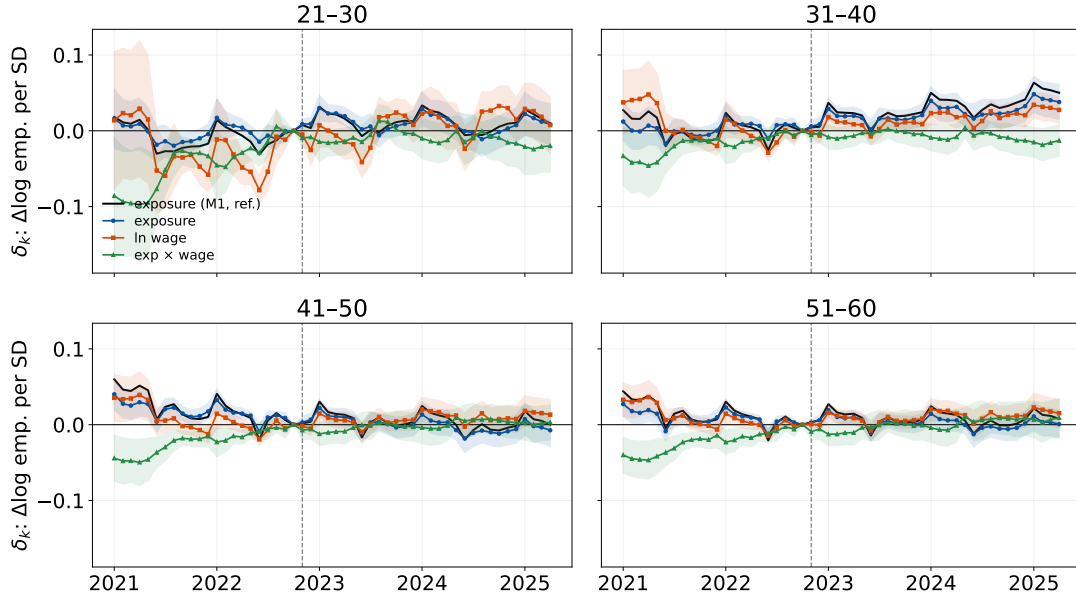
Notes: Poisson PPML (cell-level new hires = round(count $\times$ ny_jobb), private sector). Columns (1)–(3): ChatGPT window (Oct 2022 = $k=-1$, through 2025m4); (4)–(6): agentic (Apr 2025 = $k=-1$, from 2023m7). Each coefficient is the average post-period effect relative to the reference month ($k=-1$); each pre-period month enters as a separate control. M1 = exposure only, M2 = ln wage only, M3 = both + interaction. Treatments standardized: $z(\text{exp}) = \text{Eloundou } \beta$ (pooled), $z(\text{lnw}) = \ln$ full-time-equivalent wage within age group. Occupation and month fixed effects in all columns; SE clustered at occupation in parentheses; coefficients are $\Delta \log$ per 1 SD. Pseudo $R^2 \approx 0.99$ (FE-driven). * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

3 Log-employment OLS (alternative estimator)

Sections 1–2 use Poisson PPML on counts (robust to zero cells; implicitly employment-weighted). This section uses an alternative estimator: *OLS on log employment*, with occupation and month fixed effects. To keep it employment-representative — comparable to PPML — each cell is weighted by the occupation $\times$ age group’s *baseline* (pre-period) employment, and zero-employment cells are dropped (logs).

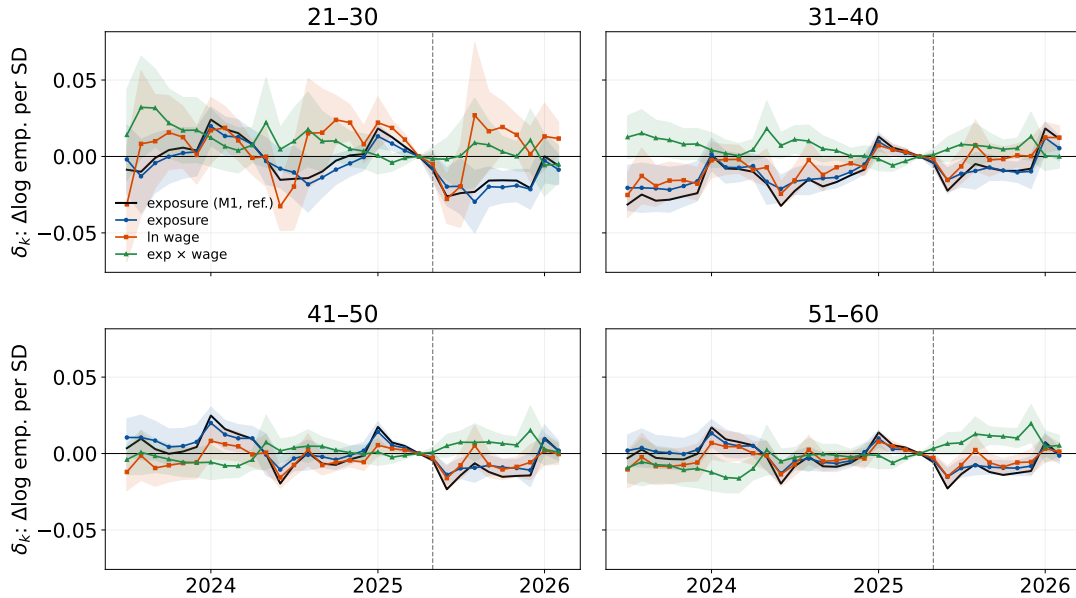
3.1 Event study

CA event study, all terms (count) [OLS log, wtd]; ref. Oct 2022 = ChatGPT



Private sector. Full model: $z(\text{exposure})$, $z(\ln \text{ FTE wage})$ and their interaction, each $\times$ event-time dummies ($k=-1$ ref). Black line = exposure estimated alone (M1). Poisson PPML, occ + month FE; 95% CI bands; SE clustered at occupation. Dashed = reference launch.

CA event study, all terms (count) [OLS log, wtd]; re-anchored Apr 2025 = agentic



Private sector. Full model: $z(\text{exposure})$, $z(\ln \text{ FTE wage})$ and their interaction, each $\times$ event-time dummies ($k=-1$ ref). Black line = exposure estimated alone (M1). Poisson PPML, occ + month FE; 95% CI bands; SE clustered at occupation. Dashed = reference launch.

Figure 3: Log employment (OLS, baseline-weighted): CA event study — ChatGPT anchor (top), agentic anchor (bottom).

3.2 Pooled post-treatment (DiD)

Table 3: AI exposure, wage, and private-sector log employment, by age group

	ChatGPT (ref. Oct 2022)			Agentic (ref. Apr 2025)		
	(1)	(2)	(3)	(4)	(5)	(6)
<i>Panel A: ages 21–30</i>						
Exposure $\times$ Post	+0.0113 (0.0072)		+0.0098 (0.0081)	-0.0156*** (0.0052)		-0.0168*** (0.0053)
ln wage $\times$ Post		+0.0120 (0.0094)	+0.0073 (0.0108)		-0.0005 (0.0086)	+0.0051 (0.0086)
Exp $\times$ wage $\times$ Post			-0.0100 (0.0109)			+0.0017 (0.0060)
Observations		19,332			11,955	
Occupations		390			387	
<i>Panel B: ages 31–40</i>						
Exposure $\times$ Post	+0.0303*** (0.0062)		+0.0236*** (0.0074)	-0.0050* (0.0030)		-0.0059* (0.0034)
ln wage $\times$ Post		+0.0270*** (0.0055)	+0.0151*** (0.0057)		-0.0020 (0.0035)	+0.0006 (0.0040)
Exp $\times$ wage $\times$ Post			-0.0070 (0.0075)			+0.0051 (0.0036)
Observations		19,427			11,962	
Occupations		387			386	
<i>Panel C: ages 41–50</i>						
Exposure $\times$ Post	+0.0045 (0.0062)		-0.0000 (0.0072)	-0.0093*** (0.0022)		-0.0062** (0.0026)
ln wage $\times$ Post		+0.0080 (0.0058)	+0.0080 (0.0066)		-0.0080*** (0.0027)	-0.0052 (0.0032)
Exp $\times$ wage $\times$ Post			-0.0007 (0.0087)			+0.0059** (0.0029)
Observations		19,360			11,896	
Occupations		387			382	
<i>Panel D: ages 51–60</i>						
Exposure $\times$ Post	+0.0078 (0.0055)		+0.0028 (0.0060)	-0.0092*** (0.0019)		-0.0068*** (0.0019)
ln wage $\times$ Post		+0.0118** (0.0053)	+0.0103* (0.0059)		-0.0068*** (0.0021)	-0.0045** (0.0022)
Exp $\times$ wage $\times$ Post			+0.0001 (0.0091)			+0.0092*** (0.0023)
Observations		19,309			11,908	
Occupations		389			386	
Occupation FE	Yes	Yes	Yes	Yes	Yes	Yes
Month FE	Yes	Yes	Yes	Yes	Yes	Yes

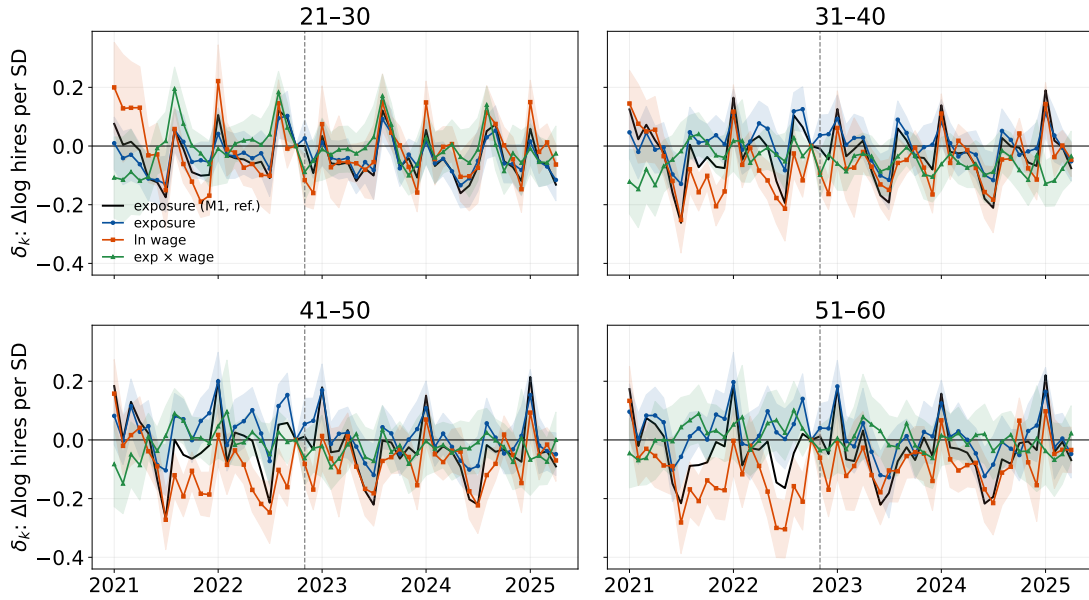
Notes: OLS on log employment, weighted by baseline (pre-period) employment (zero cells dropped) (cell-level log employment, private sector). Columns (1)–(3): ChatGPT window (Oct 2022 = $k=-1$, through 2025m4); (4)–(6): agentic (Apr 2025 = $k=-1$, from 2023m7). Each coefficient is the average post-period effect relative to the reference month ($k=-1$); each pre-period month enters as a separate control. M1 = exposure only, M2 = ln wage only, M3 = both + interaction. Treatments standardized: $z(\text{exp})$ = Eloundou β (pooled), $z(\text{lnw})$ = ln full-time-equivalent wage within age group. Occupation and month fixed effects in all columns; SE clustered at occupation in parentheses; coefficients are Δ log per 1 SD. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

4 Log new-hires OLS (alternative estimator)

The same OLS-on-logs estimator applied to new hires. Because hires are a small flow, taking logs drops cell-months with zero hires (a selected subsample); PPML in Section 2 handles these directly and is preferred for hires. Weighted by baseline (pre-period) hires.

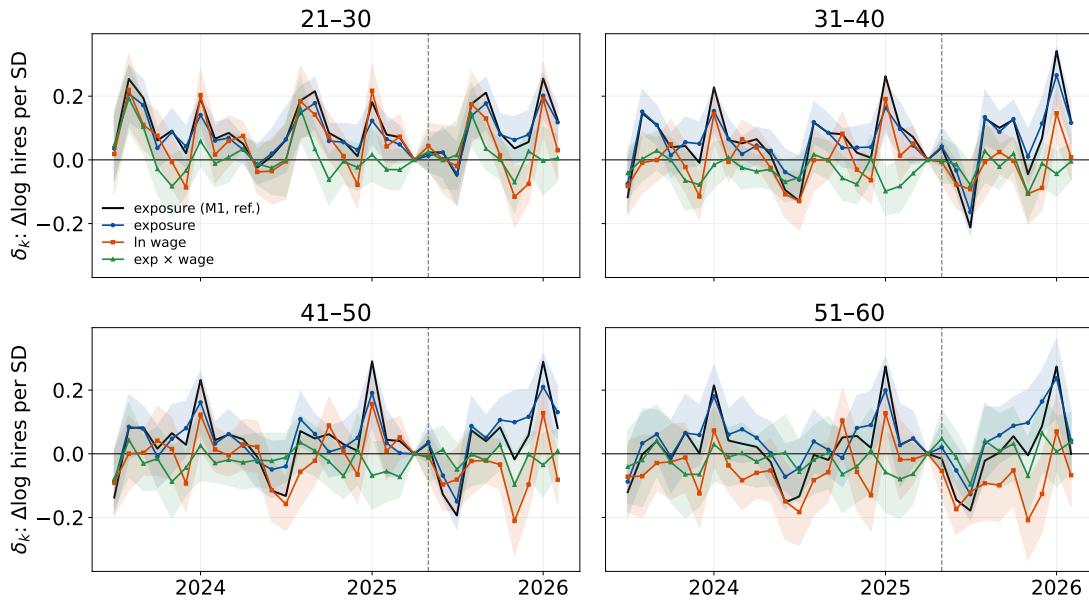
4.1 Event study

CA event study, all terms (nyjobb) [OLS log, wtd]; ref. Oct 2022 = ChatGPT



Private sector. Full model: $z(\text{exposure})$, $z(\ln \text{ FTE wage})$ and their interaction, each $\times$ event-time dummies ($k=-1$ ref). Black line = exposure estimated alone (M1). Poisson PPML, occ + month FE; 95% CI bands; SE clustered at occupation. Dashed = reference launch.

CA event study, all terms (nyjobb) [OLS log, wtd]; re-anchored Apr 2025 = agentic



Private sector. Full model: $z(\text{exposure})$, $z(\ln \text{ FTE wage})$ and their interaction, each $\times$ event-time dummies ($k=-1$ ref). Black line = exposure estimated alone (M1). Poisson PPML, occ + month FE; 95% CI bands; SE clustered at occupation. Dashed = reference launch.

Figure 4: Log new hires (OLS, baseline-weighted): CA event study — ChatGPT anchor (top), agentic anchor (bottom).

4.2 Pooled post-treatment (DiD)

Table 4: AI exposure, wage, and private-sector log new hires, by age group

	ChatGPT (ref. Oct 2022)			Agentic (ref. Apr 2025)		
	(1)	(2)	(3)	(4)	(5)	(6)
<i>Panel A: ages 21–30</i>						
Exposure $\times$ Post	-0.0465* (0.0251)		-0.0410 (0.0264)	+0.0941*** (0.0231)		+0.0850*** (0.0242)
ln wage $\times$ Post		-0.0353 (0.0215)	-0.0235 (0.0230)		+0.0613** (0.0270)	+0.0376 (0.0271)
Exp $\times$ wage $\times$ Post			-0.0155 (0.0311)			+0.0181 (0.0254)
Observations		16,781			10,229	
Occupations		363			360	
<i>Panel B: ages 31–40</i>						
Exposure $\times$ Post	-0.0279 (0.0249)		-0.0011 (0.0249)	+0.0596*** (0.0207)		+0.0703*** (0.0235)
ln wage $\times$ Post		-0.0540*** (0.0207)	-0.0532*** (0.0198)		+0.0157 (0.0236)	-0.0201 (0.0251)
Exp $\times$ wage $\times$ Post			-0.0636** (0.0311)			-0.0232 (0.0216)
Observations		16,618			10,097	
Occupations		367			359	
<i>Panel C: ages 41–50</i>						
Exposure $\times$ Post	-0.0388 (0.0247)		+0.0010 (0.0244)	+0.0321 (0.0204)		+0.0618** (0.0294)
ln wage $\times$ Post		-0.0717*** (0.0220)	-0.0723*** (0.0237)		-0.0188 (0.0248)	-0.0522* (0.0315)
Exp $\times$ wage $\times$ Post			-0.0320 (0.0347)			-0.0166 (0.0285)
Observations		15,847			9,653	
Occupations		364			359	
<i>Panel D: ages 51–60</i>						
Exposure $\times$ Post	-0.0405 (0.0258)		+0.0001 (0.0272)	+0.0057 (0.0264)		+0.0564* (0.0341)
ln wage $\times$ Post		-0.0768*** (0.0276)	-0.0767** (0.0316)		-0.0637* (0.0353)	-0.0923** (0.0413)
Exp $\times$ wage $\times$ Post			-0.0061 (0.0317)			+0.0065 (0.0397)
Observations		14,525			8,910	
Occupations		361			352	
Occupation FE	Yes	Yes	Yes	Yes	Yes	Yes
Month FE	Yes	Yes	Yes	Yes	Yes	Yes

Notes: OLS on log new hires, weighted by baseline (pre-period) new hires (zero cells dropped) (cell-level log new hires, private sector). Columns (1)–(3): ChatGPT window (Oct 2022 = $k=-1$, through 2025m4); (4)–(6): agentic (Apr 2025 = $k=-1$, from 2023m7). Each coefficient is the average post-period effect relative to the reference month ($k=-1$); each pre-period month enters as a separate control. M1 = exposure only, M2 = ln wage only, M3 = both + interaction. Treatments standardized: $z(\text{exp}) = \text{Eloundou } \beta$ (pooled), $z(\text{lnw}) = \ln$ full-time-equivalent wage within age group. Occupation and month fixed effects in all columns; SE clustered at occupation in parentheses; coefficients are $\Delta \log$ per 1 SD. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

5 Firm-FE event studies (Poisson PPML, within-firm)

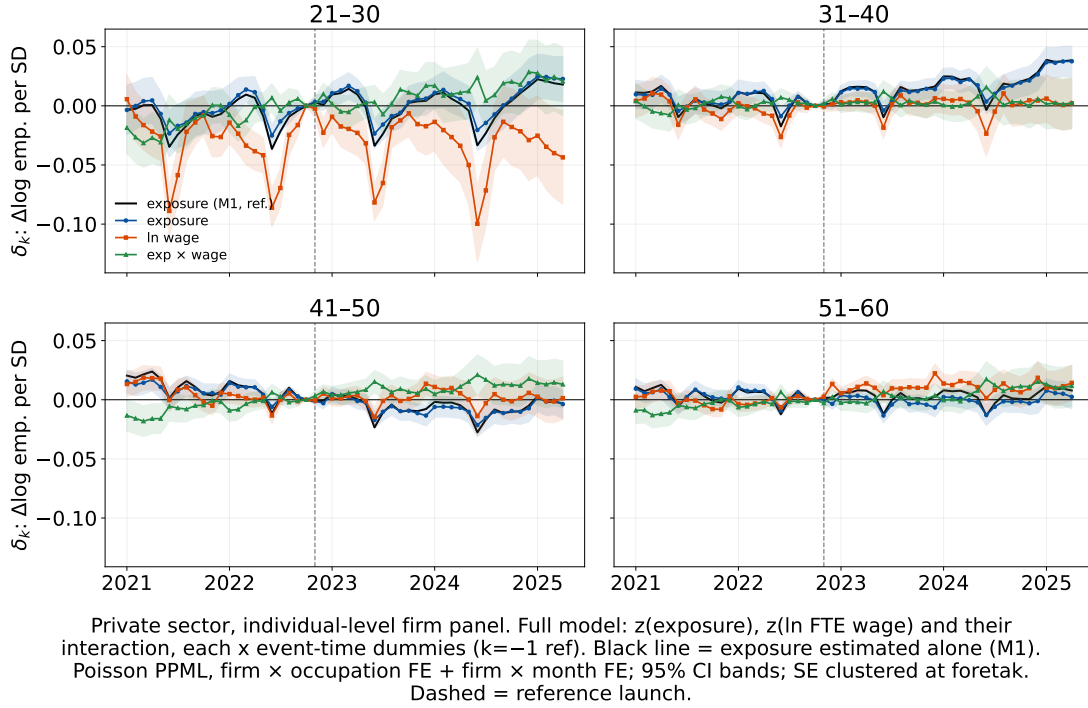
Sections 1–4 fit the CA model on the microdata.no aggregate cell panel (occupation + month FE). This chapter replicates the same specification on the secure-server individual-level firm panel and substitutes the Brynjolfsson–Chandar–Chen (BCC) eq. 4.1 fixed-effect structure: $\text{firm} \times \text{occupation} + \text{firm} \times \text{month}$. The $\text{firm} \times \text{month}$ FE absorbs every firm-level time shock (sector, geography, demand, etc.); the $\text{firm} \times \text{occupation}$ FE absorbs firm-specific occupation baselines. The treatment paths $i(k, z_{\text{exp}}) + i(k, z_{\text{wage}}) + i(k, z_{\text{exp} \times \text{wage}})$ survive the FE because they vary by occupation $\times$ time within firm-month — BCC’s within-firm differential-exposure identification.

The exposure-cell is **firm $\times$ occupation**, the continuous-treatment-faithful spec: the cell coincides with the level at which $z_{\text{exp}}/z_{\text{wage}}$ vary, so it absorbs firm-occupation baselines (subsuming a plain occupation FE). It identifies off firms with ≥ 2 occupations in the same age-month. This is the within-firm robustness check that the cell-level paths in §§1–4 are not driven by between-firm sorting on occupation.

Sample: `in_headline_priv` (private sector, ≥ 20 workers in the age window per active month), 21–30 / 31–40 / 41–50 / 51–60 decade bins. SE clustered at `foretak`. The observation counts reported in the tables are the fe pois pre-fit sample size (after singleton-FE and all-zero outcome drops that fe pois removes before estimation), so they are smaller than the raw panel counts in the $\text{firm} \times \text{month}$ panel. z_{exp} is the Eloundou GPT-4 β pooled across occupations, identical to §§1–4. z_{wage} is computed natively on the secure private sample (pre-ChatGPT employment-weighted log FTE wage per occupation $\times$ age, z-scored within age), so the wage object is the secure-sample analogue of §1’s wage rather than the identical vector.

5.1 Employment (count), firm $\times$ occupation FE

Firm-FE CA event study, all terms (count); re-anchored Oct 2022 = ChatGPT; firm $\times$ occupation FE



Firm-FE CA event study, all terms (count); re-anchored Apr 2025 = agentic; firm $\times$ occupation FE

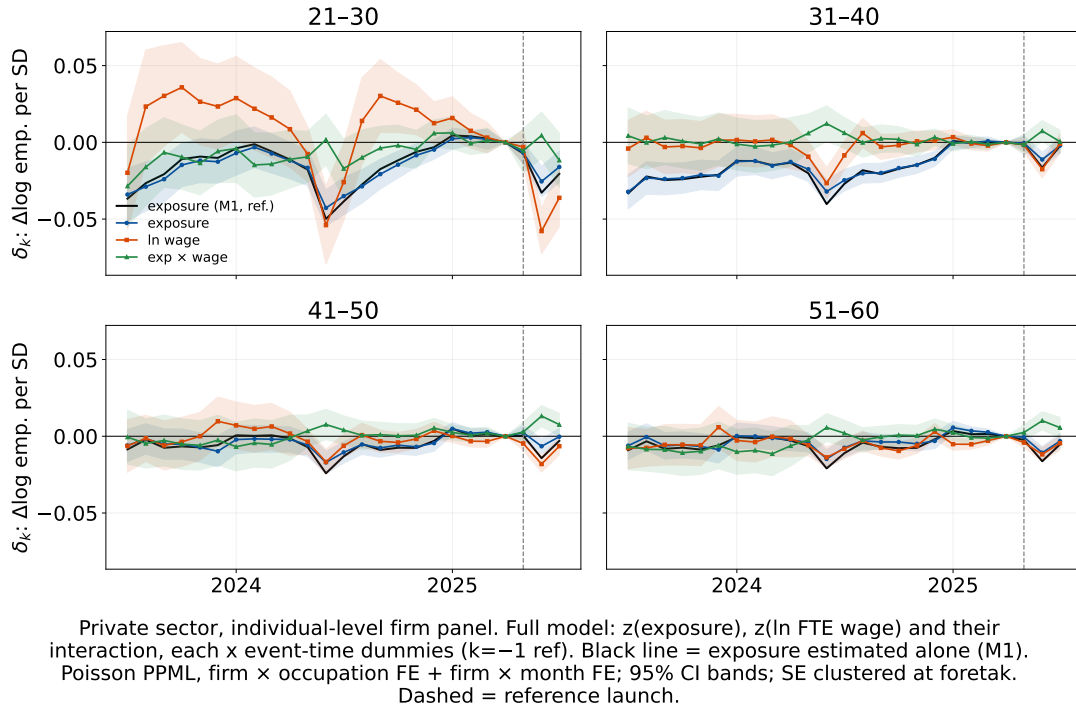
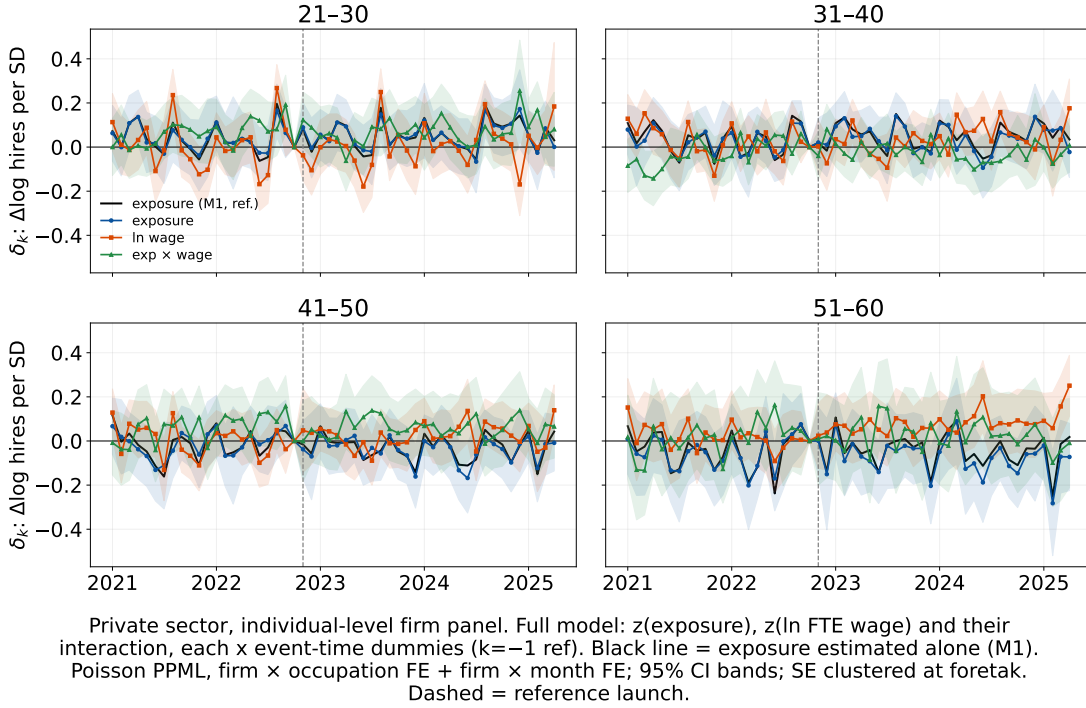


Figure 5: Firm-FE CA event study, employment (count), firm $\times$ occupation cell — ChatGPT anchor (top), agentic anchor (bottom).

5.2 New hires (nyjobb), firm $\times$ occupation FE

Firm-FE CA event study, all terms (nyjobb); re-anchored Oct 2022 = ChatGPT; firm $\times$ occupation FE



Firm-FE CA event study, all terms (nyjobb); re-anchored Apr 2025 = agentic; firm $\times$ occupation FE

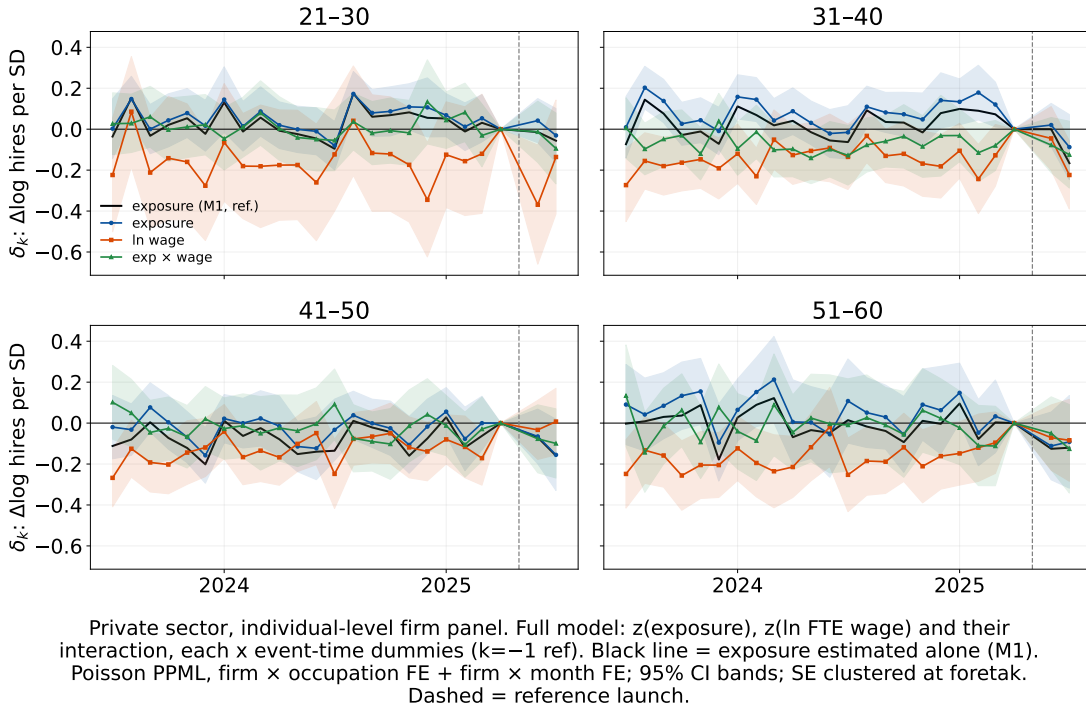


Figure 6: Firm-FE CA event study, new hires (nyjobb), firm $\times$ occupation cell — ChatGPT anchor (top), agentic anchor (bottom).

6 Firm-FE pooled DiD (Poisson PPML, within-firm)

Pooled post-treatment analogue of Section 5, mirroring Section 2’s table format. Each pre-period month enters as its own control level; all post months collapse to “POST” with the reference month ($k = -1$) omitted, so each POST coefficient is the average post-period effect relative to the reference month. Three nested specifications per (timing, age) cell:

- **M1** (*exp*): $z_{\text{exp}} \cdot \text{Post}$ only.
- **M2** (*wage*): $z_{\text{wage}} \cdot \text{Post}$ only.
- **M3** (*full*): $z_{\text{exp}} \text{Post} + z_{\text{wage}} \text{Post} + (z_{\text{exp}} z_{\text{wage}}) \text{Post}$.

6.1 Employment (count), firm $\times$ occupation FE

Table 5: Firm-FE CA pooled DiD on private-sector employment headcount, by age group (firm-occupation cell)

	ChatGPT (ref. Oct 2022)			Agentic (ref. Apr 2025)		
	(1)	(2)	(3)	(4)	(5)	(6)
<i>Panel A: ages 21–30</i>						
Exposure $\times$ Post	+0.0011 (0.0061)		+0.0048 (0.0059)	-0.0206*** (0.0038)		-0.0164*** (0.0039)
ln wage $\times$ Post		-0.0296*** (0.0107)	-0.0309*** (0.0107)		-0.0364*** (0.0062)	-0.0335*** (0.0064)
Exp $\times$ wage $\times$ Post			+0.0111 (0.0083)			-0.0036 (0.0074)
Observations		4,224,298			1,662,029	
Foretak (clusters)		20,556			17,426	
<i>Panel B: ages 31–40</i>						
Exposure $\times$ Post	+0.0169*** (0.0037)		+0.0164*** (0.0038)	-0.0068*** (0.0020)		-0.0047** (0.0022)
ln wage $\times$ Post		+0.0086* (0.0048)	+0.0010 (0.0056)		-0.0081*** (0.0019)	-0.0069*** (0.0024)
Exp $\times$ wage $\times$ Post			+0.0038 (0.0056)			+0.0023 (0.0033)
Observations		5,367,725			2,187,096	
Foretak (clusters)		20,607			17,457	
<i>Panel C: ages 41–50</i>						
Exposure $\times$ Post	-0.0061** (0.0031)		-0.0066* (0.0036)	-0.0057*** (0.0019)		-0.0016 (0.0024)
ln wage $\times$ Post		+0.0020 (0.0034)	+0.0015 (0.0046)		-0.0083*** (0.0017)	-0.0099*** (0.0023)
Exp $\times$ wage $\times$ Post			+0.0103** (0.0052)			+0.0080** (0.0032)
Observations		5,498,699			2,229,602	
Foretak (clusters)		20,482			17,361	
<i>Panel D: ages 51–60</i>						
Exposure $\times$ Post	+0.0031 (0.0028)		-0.0010 (0.0032)	-0.0081*** (0.0015)		-0.0049** (0.0019)
ln wage $\times$ Post		+0.0119*** (0.0027)	+0.0106*** (0.0037)		-0.0066*** (0.0013)	-0.0069*** (0.0019)
Exp $\times$ wage $\times$ Post			+0.0050 (0.0046)			+0.0062** (0.0027)
Observations		4,940,889			2,104,661	
Foretak (clusters)		19,992			16,980	
firm $\times$ occupation FE	Yes	Yes	Yes	Yes	Yes	Yes
Firm $\times$ month FE	Yes	Yes	Yes	Yes	Yes	Yes

Notes: Poisson PPML (`fixest::fepois`) on the individual-level firm panel (private sector, `in_headline_priv`). Outcome: cell-level employment headcount summed within (foretak, occupation, age, month). Fixed effects: firm $\times$ occupation + firm $\times$ month (BCC eq. 4.1 within-firm time-shock control). Columns (1)–(3): ChatGPT window (Oct 2022 = $k=-1$, through 2025m4); (4)–(6): agentic (Apr 2025 = $k=-1$, from 2023m7). Each coefficient is the average post-period effect relative to the reference month ($k=-1$); each pre-period month enters as a separate control. M1 = exposure only, M2 = ln wage only, M3 = both + interaction. Treatments standardized: $z(\text{exp}) = \text{Eloundou } \beta$ (pooled across occupations), $z(\text{lnw}) = \ln$ full-time-equivalent wage within age group (pre-ChatGPT, native to the secure private sample). SE clustered at foretak in parentheses; coefficients are $\Delta \log$ per 1 SD. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

6.2 New hires (nyjobb), firm $\times$ occupation FE

Table 6: Firm-FE CA pooled DiD on private-sector new hires = round(count $\times$ ny_jobb), by age group (firm-occupation cell)

	ChatGPT (ref. Oct 2022)			Agentic (ref. Apr 2025)		
	(1)	(2)	(3)	(4)	(5)	(6)
<i>Panel A: ages 21–30</i>						
Exposure $\times$ Post	+0.0588 (0.0553)		+0.0559 (0.0540)	-0.0292 (0.0437)		+0.0121 (0.0526)
ln wage $\times$ Post		+0.0356 (0.0460)	+0.0166 (0.0450)		-0.2826* (0.1443)	-0.2853** (0.1443)
Exp $\times$ wage $\times$ Post			+0.0741 (0.0546)			-0.0484 (0.0733)
Observations		904,733			281,549	
Foretak (clusters)		20,556			17,426	
<i>Panel B: ages 31–40</i>						
Exposure $\times$ Post	+0.0581 (0.0472)		+0.0480 (0.0464)	-0.0655 (0.0503)		-0.0225 (0.0524)
ln wage $\times$ Post		+0.0483 (0.0390)	+0.0344 (0.0370)		-0.1397** (0.0664)	-0.1097 (0.0730)
Exp $\times$ wage $\times$ Post			-0.0174 (0.0404)			-0.1003 (0.0637)
Observations		918,685			273,260	
Foretak (clusters)		20,607			17,457	
<i>Panel C: ages 41–50</i>						
Exposure $\times$ Post	-0.0224 (0.0484)		-0.0312 (0.0507)	-0.1044* (0.0588)		-0.1030 (0.0629)
ln wage $\times$ Post		+0.0331 (0.0413)	+0.0273 (0.0393)		-0.0694 (0.0616)	-0.0138 (0.0674)
Exp $\times$ wage $\times$ Post			+0.0575 (0.0571)			-0.0921 (0.0755)
Observations		674,351			197,171	
Foretak (clusters)		20,482			17,361	
<i>Panel D: ages 51–60</i>						
Exposure $\times$ Post	-0.0359 (0.0596)		-0.0654 (0.0569)	-0.1233 (0.0764)		-0.1054 (0.0826)
ln wage $\times$ Post		+0.0596 (0.0527)	+0.0752 (0.0471)		-0.1355** (0.0679)	-0.0749 (0.0764)
Exp $\times$ wage $\times$ Post			+0.0239 (0.0802)			-0.0819 (0.0825)
Observations		403,629			122,568	
Foretak (clusters)		19,992			16,980	
firm $\times$ occupation FE	Yes	Yes	Yes	Yes	Yes	Yes
Firm $\times$ month FE	Yes	Yes	Yes	Yes	Yes	Yes

Notes: Poisson PPML (fixest::fepois) on the individual-level firm panel (private sector, in_headline_priv). Outcome: cell-level new hires = round(count $\times$ ny_jobb) summed within (foretak, occupation, age, month). Fixed effects: firm $\times$ occupation + firm $\times$ month (BCC eq. 4.1 within-firm time-shock control). Columns (1)–(3): ChatGPT window (Oct 2022 = $k=-1$, through 2025m4); (4)–(6): agentic (Apr 2025 = $k=-1$, from 2023m7). Each coefficient is the average post-period effect relative to the reference month ($k=-1$); each pre-period month enters as a separate control. M1 = exposure only, M2 = ln wage only, M3 = both + interaction. Treatments standardized: $z(\text{exp}) = \text{Eloundou } \beta$ (pooled across occupations), $z(\text{lnw}) = \text{ln full-time-equivalent wage within age group (pre-ChatGPT, native to the secure private sample)}$. SE clustered at foretak in parentheses; coefficients are $\Delta \log$ per 1 SD. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

7 Yagan-style event studies (microdata.no, linear OLS)

Sections 1–6 indexed every series to a single reference month ($k = -1$, i.e. either Oct 2022 for ChatGPT or Apr 2025 for the agentic anchor). This chapter replaces that single-month index with the Yagan-(2020)-style ratio against a *12-month pre-window* mean baseline. For each cell $c = (\text{yrke4}, \text{alder_gr})$ in the private sector and each anchor, with $\bar{y}_c$ the mean of y over $k \in [-12, -1]$:

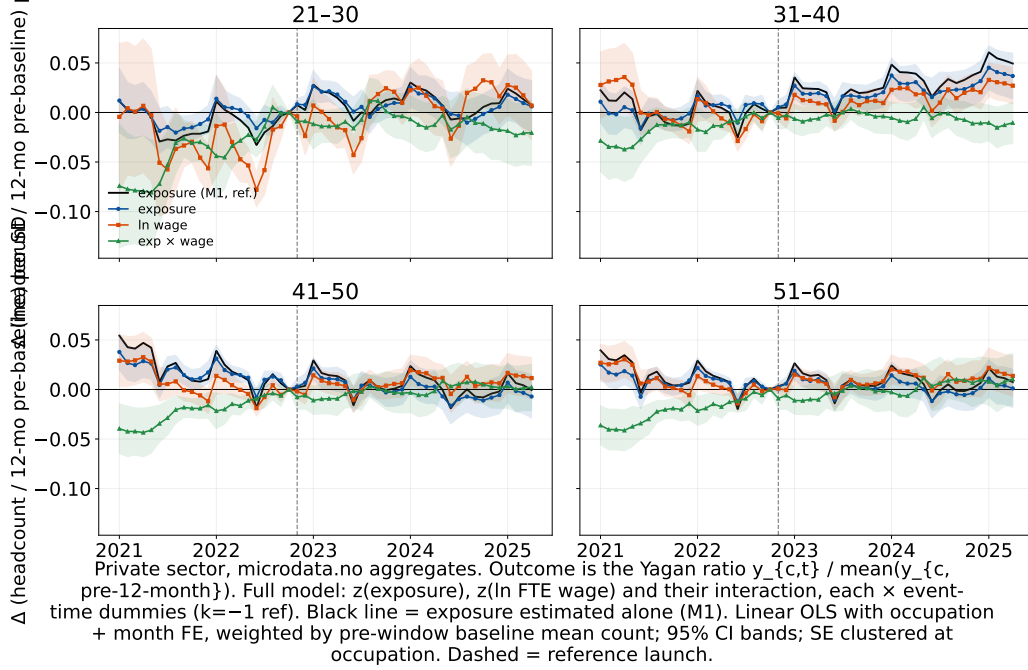
$$y_{c,t}^{\text{ratio}} = \frac{y_{c,t}}{\bar{y}_c} \quad (1 = \text{baseline}, 1.05 = +5\% \text{ above baseline}).$$

Pre-windows: *chatgpt* = 2021m11–2022m10, *agentic* = 2024m05–2025m04. The 12-month window gives a smoother, less anomaly-prone *denominator* than picking the single reference month $k = -1$. It does *not* absorb seasonality, however: the denominator $\bar{y}_c$ is a scalar per cell, so any calendar-month pattern in the numerator $y_{c,t}$ propagates directly into $y_{c,t}^{\text{ratio}}$ (a high-employment month divided by the year’s mean is still high). Sections 9–10 implement a same-month-of-year ratio that does remove seasonality. The current ratio is the cell-level analog of Yagan’s Fig. 6B individual-level $\text{EARNINGS}_{i,2015}/\overline{\text{EARNINGS}}_{i,\text{pre}}$ construction.

Cells with $\bar{y}_c = 0$ are dropped (the ratio is undefined and feols rejects zero weights). Linear OLS (`fixest::feols`) on y^{ratio} with $z(\text{exp}) \times \text{event-time}$, $z(\text{lnw}) \times \text{event-time}$ and their three-way interaction. Cells are weighted by the same $\bar{y}_c$ that defines the denominator, so each cell-month contributes in proportion to the cell’s pre-window average headcount: a 5000-worker cell counts 500 times more than a 10-worker cell. This is the cell-aggregate analog of the equal-weight-per-individual convention in Yagan’s regressions and is what makes the recovered coefficients an employment-weighted average ratio change per SD of treatment. Occupation and month FE; SE clustered at occupation. Same FE structure and identifying variation (within-occupation $\times$ time interactions with the standardized treatments) as Section 1; only the estimator (OLS vs. PPML) and the outcome (y^{ratio} vs. raw count y) differ.

7.1 Employment (count)

CA event study, all terms (employment headcount); Yagan ratio ($y / 12\text{-month pre-baseline}$); re-anchored Oct 2022 = ChatGPT



CA event study, all terms (employment headcount); Yagan ratio ($y / 12\text{-month pre-baseline}$); re-anchored Apr 2025 = agentic

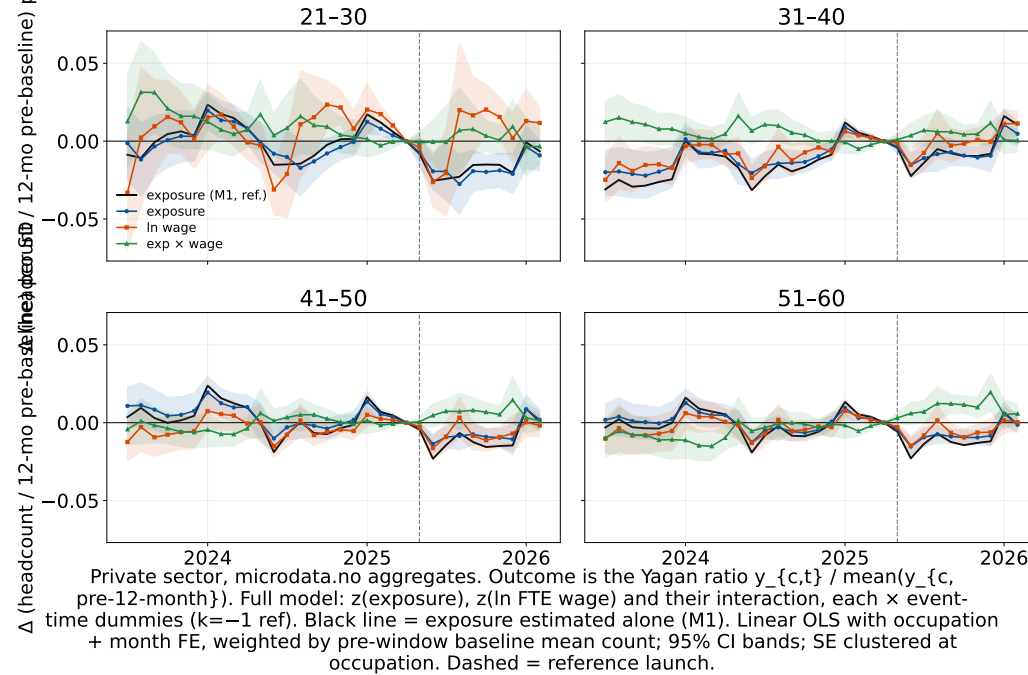
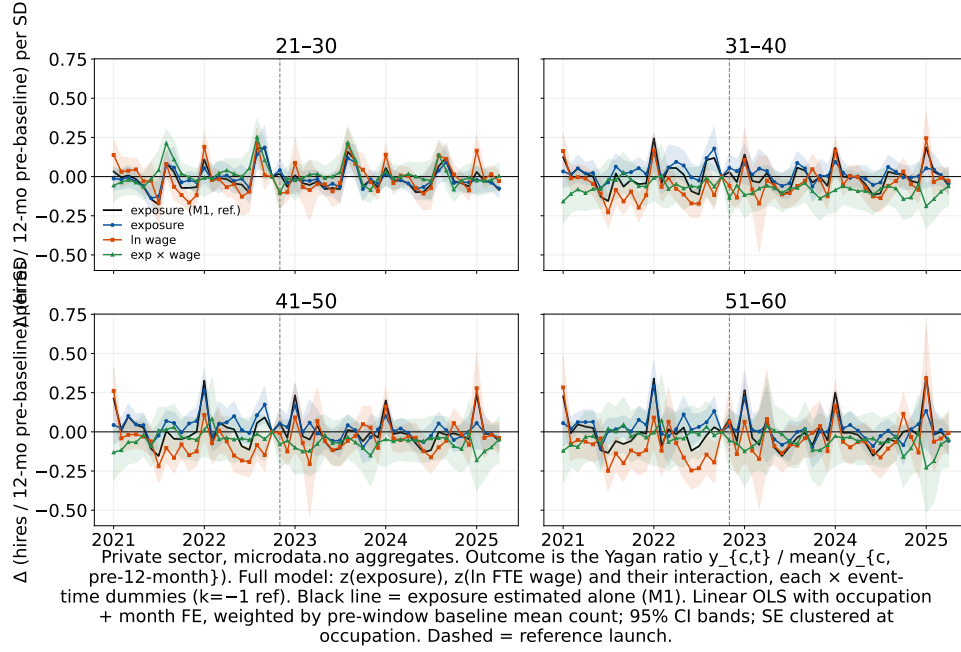


Figure 7: Yagan-ratio CA event study, employment headcount — ChatGPT anchor (top), agentic anchor (bottom). Coefficients are $\Delta(y/\bar{y}_c^{\text{pre}})$ per SD.

7.2 New hires (nyjobb)

CA event study, all terms (new hires (round count $\times$ ny_jobb)); Yagan ratio (y / 12-month pre-baseline); re-anchored Oct 2022 = ChatGPT



CA event study, all terms (new hires (round count $\times$ ny_jobb)); Yagan ratio (y / 12-month pre-baseline); re-anchored Apr 2025 = agentic

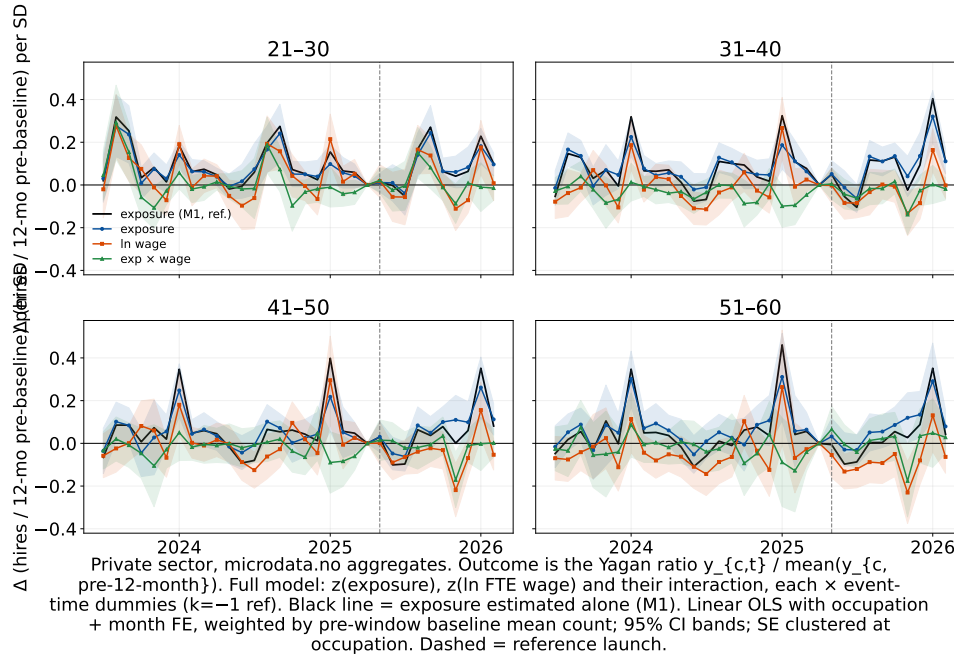


Figure 8: Yagan-ratio CA event study, new hires — ChatGPT anchor (top), agentic anchor (bottom). Coefficients are $\Delta(y/\bar{y}_c^{\text{pre}})$ per SD.

8 Yagan-style pooled DiD (microdata.no, linear OLS)

Pooled-DiD counterpart to Section 7, mirroring Section 2’s table format. Each pre-period month enters as its own control level; all post months collapse to “POST” with the reference month ($k = -1$) omitted, so each POST coefficient is the average post-period effect on the Yagan ratio. M1 = exposure only, M2 = ln wage only, M3 = both + interaction.

8.1 Employment (count)

Table 7: Yagan-style CA pooled DiD on private-sector employment headcount (Yagan ratio ($y_{c,t} / \bar{y}_{c,\text{pre-12mo}}$)), by age group

	ChatGPT (ref. Oct 2022)			Agentic (ref. Apr 2025)		
	(1)	(2)	(3)	(4)	(5)	(6)
<i>Panel A: ages 21–30</i>						
Exposure $\times$ Post	+0.0101 (0.0070)		+0.0087 (0.0078)	-0.0154*** (0.0053)		-0.0165*** (0.0054)
ln wage $\times$ Post		+0.0110 (0.0086)	+0.0064 (0.0101)		-0.0005 (0.0083)	+0.0048 (0.0082)
Exp $\times$ wage $\times$ Post			-0.0103 (0.0104)			+0.0019 (0.0058)
Observations		20,124			12,320	
Occupations		387			385	
<i>Panel B: ages 31–40</i>						
Exposure $\times$ Post	+0.0299*** (0.0062)		+0.0231*** (0.0073)	-0.0054* (0.0031)		-0.0059* (0.0034)
ln wage $\times$ Post		+0.0263*** (0.0054)	+0.0146** (0.0057)		-0.0026 (0.0036)	-0.0000 (0.0039)
Exp $\times$ wage $\times$ Post			-0.0055 (0.0073)			+0.0045 (0.0035)
Observations		19,968			12,288	
Occupations		384			384	
<i>Panel C: ages 41–50</i>						
Exposure $\times$ Post	+0.0043 (0.0061)		+0.0003 (0.0072)	-0.0096*** (0.0023)		-0.0060** (0.0026)
ln wage $\times$ Post		+0.0071 (0.0058)	+0.0071 (0.0067)		-0.0090*** (0.0029)	-0.0062* (0.0033)
Exp $\times$ wage $\times$ Post			-0.0011 (0.0087)			+0.0059** (0.0030)
Observations		20,020			12,256	
Occupations		385			383	
<i>Panel D: ages 51–60</i>						
Exposure $\times$ Post	+0.0078 (0.0056)		+0.0031 (0.0061)	-0.0096*** (0.0019)		-0.0068*** (0.0019)
ln wage $\times$ Post		+0.0113** (0.0055)	+0.0096 (0.0062)		-0.0074*** (0.0022)	-0.0051** (0.0023)
Exp $\times$ wage $\times$ Post			+0.0006 (0.0092)			+0.0093*** (0.0024)
Observations		20,176			12,288	
Occupations		388			384	
Occupation FE	Yes	Yes	Yes	Yes	Yes	Yes
Month FE	Yes	Yes	Yes	Yes	Yes	Yes

Notes: Linear OLS (**fixest::feols**) on the microdata.no cell aggregates (private sector, $\text{sektor} = 2$). Outcome is the Yagan ratio $y_{c,t} / \bar{y}_{c,\text{pre}}$ where the baseline is the mean of y over the 12-month pre-window $k \in [-12, -1]$. Columns (1)–(3): ChatGPT window (Oct 2022 = $k = -1$, through 2025m4); (4)–(6): agentic (Apr 2025 = $k = -1$, from 2023m7). Each coefficient is the average post-period effect relative to the reference month ($k = -1$); each pre-period month enters as a separate control. M1 = exposure only, M2 = ln wage only, M3 = both + interaction. Treatments standardized: $z(\text{exp}) = \text{Eloundou } \beta$ (pooled), $z(\text{lnw}) = \ln$ full-time-equivalent wage (pre-ChatGPT, within age group). Cells weighted by the pre-window baseline mean count. Occupation + month FE; SE clustered at occupation in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

8.2 New hires (nyjobb)

Table 8: Yagan-style CA pooled DiD on private-sector new hires = $\text{round}(\text{count} \times \text{ny_jobb})$ (Yagan ratio $(y_{c,t} / \bar{y}_{c,\text{pre}-12\text{mo}})$), by age group

	ChatGPT (ref. Oct 2022)			Agentic (ref. Apr 2025)		
	(1)	(2)	(3)	(4)	(5)	(6)
<i>Panel A: ages 21–30</i>						
Exposure $\times$ Post	-0.0213 (0.0233)		-0.0195 (0.0256)	+0.0894*** (0.0237)		+0.0853*** (0.0257)
ln wage $\times$ Post		-0.0131 (0.0219)	-0.0055 (0.0251)		+0.0450 (0.0305)	+0.0205 (0.0302)
Exp $\times$ wage $\times$ Post			+0.0022 (0.0343)			+0.0113 (0.0234)
Observations		18,668			11,392	
Occupations		359			356	
<i>Panel B: ages 31–40</i>						
Exposure $\times$ Post	-0.0036 (0.0350)		+0.0123 (0.0307)	+0.0833*** (0.0224)		+0.0962*** (0.0251)
ln wage $\times$ Post		-0.0288 (0.0366)	-0.0354 (0.0345)		+0.0207 (0.0249)	-0.0265 (0.0255)
Exp $\times$ wage $\times$ Post			-0.0937 (0.0573)			-0.0296 (0.0211)
Observations		18,980			11,392	
Occupations		365			356	
<i>Panel C: ages 41–50</i>						
Exposure $\times$ Post	-0.0085 (0.0359)		+0.0017 (0.0267)	+0.0504** (0.0196)		+0.0737** (0.0301)
ln wage $\times$ Post		-0.0300 (0.0394)	-0.0310 (0.0409)		-0.0035 (0.0268)	-0.0432 (0.0344)
Exp $\times$ wage $\times$ Post			-0.0746 (0.0664)			-0.0166 (0.0308)
Observations		18,772			11,488	
Occupations		361			359	
<i>Panel D: ages 51–60</i>						
Exposure $\times$ Post	-0.0060 (0.0376)		+0.0051 (0.0296)	+0.0380 (0.0253)		+0.0789** (0.0346)
ln wage $\times$ Post		-0.0372 (0.0483)	-0.0387 (0.0545)		-0.0401 (0.0349)	-0.0779* (0.0406)
Exp $\times$ wage $\times$ Post			-0.0678 (0.0764)			+0.0025 (0.0409)
Observations		18,564			11,104	
Occupations		357			347	
Occupation FE	Yes	Yes	Yes	Yes	Yes	Yes
Month FE	Yes	Yes	Yes	Yes	Yes	Yes

Notes: Linear OLS (`fixest::feols`) on the microdata.no cell aggregates (private sector, `sektor = 2`). Outcome is the Yagan ratio $y_{c,t} / \bar{y}_{c,\text{pre}}$ where the baseline is the mean of y over the 12-month pre-window $k \in [-12, -1]$. Columns (1)–(3): ChatGPT window (Oct 2022 = $k=-1$, through 2025m4); (4)–(6): agentic (Apr 2025 = $k=-1$, from 2023m7). Each coefficient is the average post-period effect relative to the reference month ($k=-1$); each pre-period month enters as a separate control. M1 = exposure only, M2 = ln wage only, M3 = both + interaction. Treatments standardized: $z(\text{exp}) = \text{Eloundou } \beta$ (pooled), $z(\text{lnw}) = \ln$ full-time-equivalent wage (pre-ChatGPT, within age group). Cells weighted by the pre-window baseline mean count. Occupation + month FE; SE clustered at occupation in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

9 Robustness: YoY-matched event studies (microdata.no, linear OLS)

Sections 7–8 used the *mean* of y over the 12-month pre-window as the baseline for each cell. This chapter is a robustness check that replaces that pooled-mean baseline with a *same-month-of-year* match inside the same 12-month pre-window. Concretely, for each cell c and post-period month t , the baseline is the unique pre-window observation $y_{c,m(t)}$ where $m(t)$ has the same month-of-year as t :

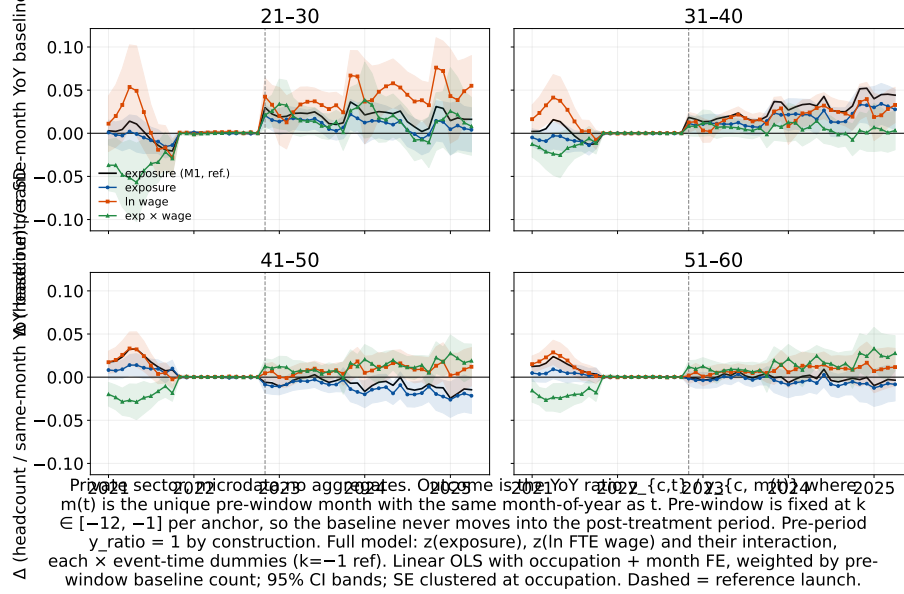
$$y_{c,t}^{\text{YoY}} = \frac{y_{c,t}}{y_{c,m(t)}},$$

with $m(t) \in \{2021\text{m}11, \dots, 2022\text{m}10\}$ for ChatGPT and $\{2024\text{m}05, \dots, 2025\text{m}04\}$ for agentic.

This strips out seasonality (sysselsetting, hires and wages all have strong calendar-month patterns) without leaving the pre-treatment window: the baseline is *anchored* at the 12-month pre-window and reused each year, so it never gets contaminated by post-treatment observations even at $k \geq 12$. By construction $y^{\text{YoY}} = 1$ for every $k \in [-12, -1]$ (numerator and denominator are the same observation), so pre-period event-time coefficients inside the baseline window are mechanically zero — there is no cross-occupation variation in the LHS to identify off. Pre-trend tests therefore have to use $k < -12$, where the baseline lies twelve months ahead of the data month. Cells with baseline $y = 0$ are dropped (ratio undefined; feols rejects zero weights). Estimator, FE structure, weights and cluster are identical to Section 7 (linear OLS, occupation + month FE, cells weighted by the pre-window baseline count, SE clustered at occupation); only the construction of the ratio’s denominator differs.

9.1 Employment (count)

CA event study, all terms (employment headcount); YoY-matched ratio (y / same-month-of-year pre-baseline); re-anchored Oct 2022 = ChatGPT



CA event study, all terms (employment headcount); YoY-matched ratio (y / same-month-of-year pre-baseline); re-anchored Apr 2025 = agentic

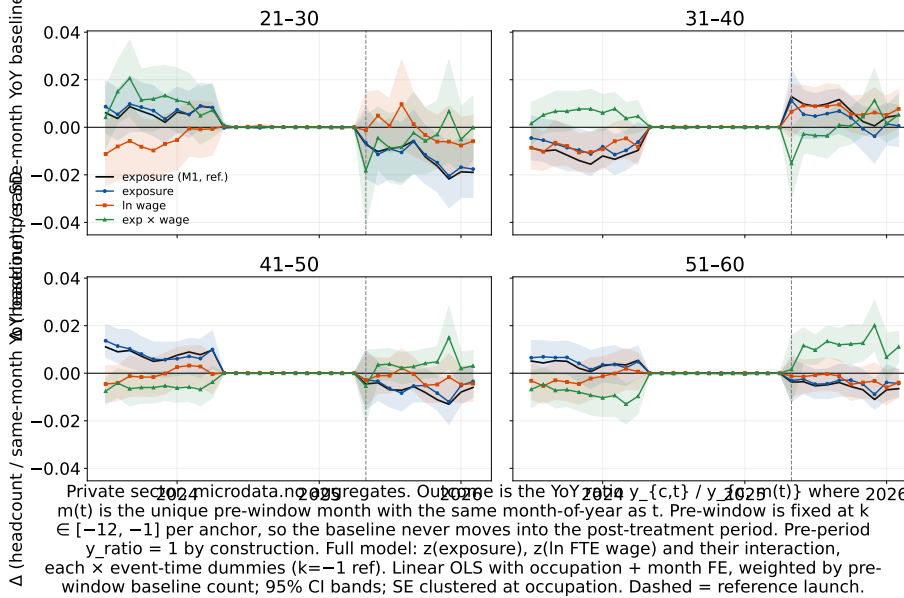
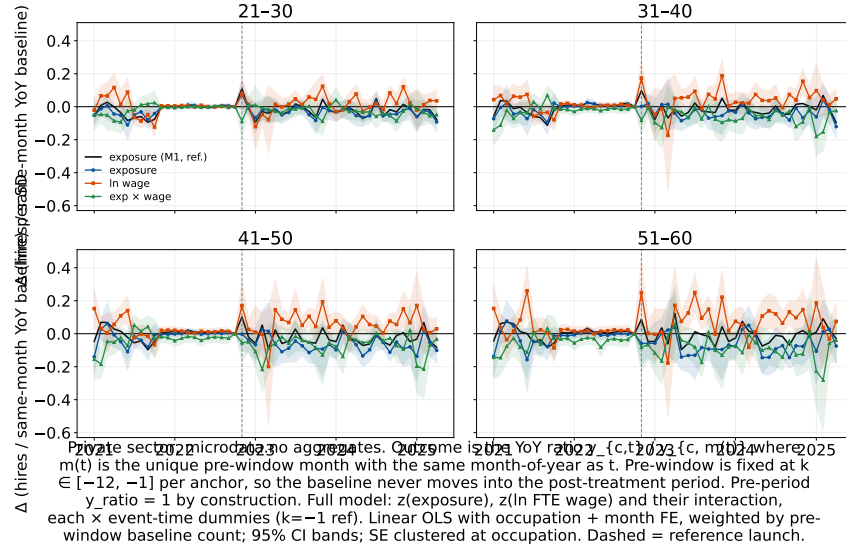


Figure 9: YoY-matched CA event study, employment headcount — ChatGPT anchor (top), agentic anchor (bottom). Coefficients are $\Delta(y_{c,t}/y_{c,m(t)})$ per SD.

9.2 New hires (nyjobb)

CA event study, all terms (new hires (round count $\times$ ny_jobb)); YoY-matched ratio (y / same-month-of-year pre-baseline); re-anchored Oct 2022 = ChatGPT



CA event study, all terms (new hires (round count $\times$ ny_jobb)); YoY-matched ratio (y / same-month-of-year pre-baseline); re-anchored Apr 2025 = agentic

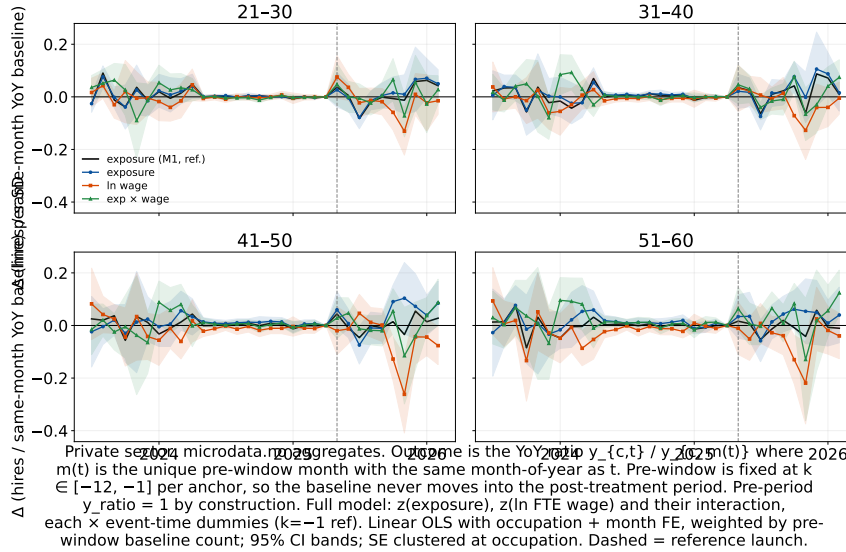


Figure 10: YoY-matched CA event study, new hires — ChatGPT anchor (top), agentic anchor (bottom). Coefficients are $\Delta(y_{c,t}/y_{c,m(t)})$ per SD.

10 Robustness: YoY-matched pooled DiD (microdata.no, linear OLS)

Pooled-DiD counterpart to Section 9, mirroring Section 8’s table format. Same kk encoding as Sections 2 and 8: each pre-period month is its own control level, all post months collapse to “POST” with the reference month ($k = -1$) omitted. The POST coefficient is the average post-period effect on the YoY ratio.

10.1 Employment (count)

Table 9: YoY-matched CA pooled DiD on private-sector employment headcount (YoY ratio $(y_{c,t}/y_{c,m(t)}^{\text{pre}})$), by age group

	ChatGPT (ref. Oct 2022)			Agentic (ref. Apr 2025)		
	(1)	(2)	(3)	(4)	(5)	(6)
<i>Panel A: ages 21–30</i>						
Exposure $\times$ Post	+0.0199** (0.0079)		+0.0105 (0.0081)	-0.0128** (0.0062)		-0.0124** (0.0061)
ln wage $\times$ Post		+0.0429*** (0.0117)	+0.0422*** (0.0131)		-0.0041 (0.0059)	-0.0013 (0.0053)
Exp $\times$ wage $\times$ Post			+0.0154 (0.0119)			-0.0050 (0.0068)
Observations		19,288			11,954	
Occupations		387			385	
<i>Panel B: ages 31–40</i>						
Exposure $\times$ Post	+0.0288*** (0.0070)		+0.0183*** (0.0069)	+0.0072* (0.0040)		+0.0035 (0.0044)
ln wage $\times$ Post		+0.0304*** (0.0067)	+0.0204*** (0.0064)		+0.0092*** (0.0035)	+0.0073** (0.0037)
Exp $\times$ wage $\times$ Post			+0.0054 (0.0073)			-0.0001 (0.0038)
Observations		19,410			11,950	
Occupations		384			384	
<i>Panel C: ages 41–50</i>						
Exposure $\times$ Post	-0.0084 (0.0062)		-0.0133** (0.0064)	-0.0075** (0.0032)		-0.0061* (0.0034)
ln wage $\times$ Post		+0.0024 (0.0066)	+0.0085 (0.0070)		-0.0054* (0.0033)	-0.0024 (0.0034)
Exp $\times$ wage $\times$ Post			+0.0126** (0.0063)			+0.0036 (0.0029)
Observations		19,404			11,885	
Occupations		385			383	
<i>Panel D: ages 51–60</i>						
Exposure $\times$ Post	-0.0009 (0.0061)		-0.0046 (0.0064)	-0.0059** (0.0029)		-0.0042 (0.0030)
ln wage $\times$ Post		+0.0080 (0.0062)	+0.0085 (0.0068)		-0.0035 (0.0030)	-0.0027 (0.0031)
Exp $\times$ wage $\times$ Post			+0.0147** (0.0070)			+0.0112*** (0.0034)
Observations		19,289			11,875	
Occupations		388			384	
Occupation FE	Yes	Yes	Yes	Yes	Yes	Yes
Month FE	Yes	Yes	Yes	Yes	Yes	Yes

Notes: Linear OLS (`fixest::feols`) on the microdata.no cell aggregates (private sector, `sektor = 2`). Outcome is the YoY ratio $y_{c,t}/y_{c,m(t)}$ where $m(t)$ is the unique pre-window month (within $k \in [-12, -1]$ per anchor) with the same month-of-year as t . The pre-window is fixed at $k \in [-12, -1]$, so the baseline never enters the post-treatment region even for $k \geq 12$. Columns (1)–(3): ChatGPT window (Oct 2022 = $k=-1$, through 2025m4); (4)–(6): agentic (Apr 2025 = $k=-1$, from 2023m7). Each coefficient is the average post-period effect on the YoY ratio relative to the reference month ($k=-1$). M1 = exposure only, M2 = ln wage only, M3 = both + interaction. Treatments standardized: $z(\text{exp}) = \text{Eloundou } \beta$ (pooled), $z(\text{lnw}) = \ln$ full-time-equivalent wage within age group. Cells weighted by the pre-window baseline count. SE clustered at occupation in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

10.2 New hires (nyjobb)

Table 10: YoY-matched CA pooled DiD on private-sector new hires = round(count $\times$ ny_jobb) (YoY ratio ($y_{c,t}/y_{c,m(t)}^{\text{pre}}$)), by age group

	ChatGPT (ref. Oct 2022)			Agentic (ref. Apr 2025)		
	(1)	(2)	(3)	(4)	(5)	(6)
<i>Panel A: ages 21–30</i>						
Exposure $\times$ Post	-0.0238* (0.0126)		-0.0314** (0.0125)	+0.0032 (0.0104)		+0.0083 (0.0112)
ln wage $\times$ Post		+0.0121 (0.0142)	+0.0177 (0.0140)		-0.0160 (0.0126)	-0.0163 (0.0137)
Exp $\times$ wage $\times$ Post			-0.0311** (0.0126)			+0.0050 (0.0128)
Observations		17,015			10,262	
Occupations		359			356	
<i>Panel B: ages 31–40</i>						
Exposure $\times$ Post	-0.0170 (0.0168)		-0.0397** (0.0161)	+0.0217* (0.0120)		+0.0331** (0.0131)
ln wage $\times$ Post		+0.0203 (0.0174)	+0.0422** (0.0167)		-0.0058 (0.0154)	-0.0210 (0.0160)
Exp $\times$ wage $\times$ Post			-0.0658*** (0.0182)			+0.0169 (0.0137)
Observations		16,785			10,119	
Occupations		365			356	
<i>Panel C: ages 41–50</i>						
Exposure $\times$ Post	-0.0265 (0.0187)		-0.0679*** (0.0161)	+0.0072 (0.0122)		+0.0397** (0.0154)
ln wage $\times$ Post		+0.0235 (0.0200)	+0.0604*** (0.0182)		-0.0328* (0.0178)	-0.0535*** (0.0200)
Exp $\times$ wage $\times$ Post			-0.0753*** (0.0216)			+0.0084 (0.0148)
Observations		16,000			9,707	
Occupations		361			359	
<i>Panel D: ages 51–60</i>						
Exposure $\times$ Post	-0.0105 (0.0207)		-0.0618*** (0.0177)	-0.0020 (0.0135)		+0.0286 (0.0177)
ln wage $\times$ Post		+0.0396 (0.0243)	+0.0714*** (0.0230)		-0.0334* (0.0193)	-0.0461** (0.0225)
Exp $\times$ wage $\times$ Post			-0.0895*** (0.0263)			+0.0349 (0.0235)
Observations		14,746			8,938	
Occupations		357			347	
Occupation FE	Yes	Yes	Yes	Yes	Yes	Yes
Month FE	Yes	Yes	Yes	Yes	Yes	Yes

Notes: Linear OLS (fixest::feols) on the microdata.no cell aggregates (private sector, sektor = 2). Outcome is the YoY ratio $y_{c,t}/y_{c,m(t)}$ where $m(t)$ is the unique pre-window month (within $k \in [-12, -1]$ per anchor) with the same month-of-year as t . The pre-window is fixed at $k \in [-12, -1]$, so the baseline never enters the post-treatment region even for $k \geq 12$. Columns (1)–(3): ChatGPT window (Oct 2022 = $k=-1$, through 2025m4); (4)–(6): agentic (Apr 2025 = $k=-1$, from 2023m7). Each coefficient is the average post-period effect on the YoY ratio relative to the reference month ($k=-1$). M1 = exposure only, M2 = ln wage only, M3 = both + interaction. Treatments standardized: $z(\text{exp}) = \text{Eloundou } \beta$ (pooled), $z(\text{lnw}) = \text{ln full-time-equivalent wage within age group}$. Cells weighted by the pre-window baseline count. SE clustered at occupation in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

References

- LORV (2026)** Lindenlaub, I., R. Oh, M. A. Rodríguez & L. Veldkamp. “Beyond Exposure: Predicting AI Adoption Based on Comparative Advantage.” Working paper, May 22, 2026. Documents the gap between AI exposure and observed adoption in German DiWaBe-IEB linked data, derives the task-level adoption condition in equation (1), and shows that a comparative-advantage index accounts for 60% of cross-occupation adoption variation versus 14% for exposure alone.
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